



THE COPPERBELT UNIVERSITY
MCS SCHOOL OF MEDICINE

Test 1 – Human Physiology. Topics: Cell, Membrane Transports, Body Fluids, Nerve-Muscle

Total Marks: 100

Duration: 2 Hours

SIN Number: _____



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Section A: Multiple Choice Questions (MCQs)

Instruction: Answer all questions on this same page. Marks 6

1. This is of primary importance in the long-term regulation of ECF volume:
A. ECF osmolarity
B. ICF osmolality
C. Nephrons
D. Medullary osmolality
E. Bone density
2. ECF osmolarity is maintained mainly by.....balance.
A. NaCl
B. KCl
C. MgSO₄
D. Ca²⁺
E. Fe⁺
3. Interstitial fluid volume is calculated using the formula:
A. ECF + plasma volume
B. ECF - plasma volume
C. TBW - ECF
D. TBW + ECF
E. TBW + plasma volume
4. The molecular weight of a solute (in grams) divided by the number of ions or particles into which it dissociates in solution is called:
A. Millimole
B. Osmole
C. Osmolality
D. Osmolarity
E. Polarity
5. _____ is a measure of the osmoles of solute per liter of solution.
A. Osmolality
B. Osmolarity
C. Polarity
D. Parity
E. Extrapolation

6. Because of the Donnan effect, cells will:
- A. Swell and burst
 - B. Shrink
 - C. No change
 - D. Become circular
 - E. Become sickled

SECTION B – MTF (Mark True/False for Each Option)

Marks 16

Instructions: For each question below, indicate whether each statement (A–D) is **True (T)** or **False (F)**.

1. **Functions of water in the body:**
 - A. Temperature regulation
 - B. Transport
 - C. Solvent
 - D. Variety

2. **Following are transcellular fluids:**
 - A. Water
 - B. Pleural fluid
 - C. Peritoneal fluid
 - D. Synovial fluid

3. **Criteria for a suitable dye in body fluid measurements:**
 - A. Must mix evenly throughout the compartment
 - B. Non-toxic
 - C. Must have no significant effect of its own
 - D. Should be relatively easy to measure

4. **Following are isotonic fluids used for IV infusion in intensive care:**
 - A. 0.9% NaCl
 - B. 5% Dextrose
 - C. Ringer's Lactate
 - D. 9% NaCl

5. **As a medical student, you are to order a solution in the emergency department based on the particular patient's:**
 - A. Serum electrolyte values
 - B. Fluid-volume balance
 - C. Economic background
 - D. Altitude

6. **Function of the sodium-potassium pump that saves the cell from swelling:**
 - A. Na^+ pumped out
 - B. Cl^- follows Na^+
 - C. H_2O follows NaCl
 - D. Ca^{2+} pumped out

7. **Example of integral proteins of the cell membrane:**
 - A. Carriers
 - B. Channels
 - C. Receptors
 - D. Antigens

8. **Plasma Membrane Action as a semi-permeable barrier aid in:**
 - A. Maintaining the difference in composition of ICF & ECF, and fluids in various organelles
 - B. Protecting the cell from toxic substances
 - C. Excretion of waste products
 - D. Transport of nutrients

SECTION C – Structured Questions

Instructions: Answer **any SEVEN (7)** questions in the white sheets provided.

Marks: 35

1. Explain the operational effect of **anaesthetics** on nerve physiology, with examples.
2. Describe the functions of **integrins** in the cell and **tight junctions** between cells.
3. Elucidate the structural design, with a labelled diagram, of **nexus** in body cells and write a note on their physiological significance.
4. In the membrane of **cardiac muscle**, intracellular **Ca²⁺** is exchanged for extracellular **Na⁺**. Explain this type of transport and give **three (3)** other relevant examples.
5. Explain **phagocytosis** with a labelled diagram.
6. Draw a labelled **concept diagram of the Donnan effect** and explain its significance.
7. Elucidate the concepts of **IPSP**, **EPSP**, and **depolarization** in a nerve membrane.
8. Describe the **pathophysiology of diarrhea in cholera disease**, with examples of **IV fluids used at Ndola Hitech Hospital**.

SECTION D – Applied Questions

Instructions: Answer **any FOUR (4)** questions in the white sheets provided.

Marks: $4 \times 10 = 40$

1. A medical science student, **Ms. Aniya**, who has been experiencing persistent insomnia for several weeks. Her family doctor prescribes a sleeping pill known to **enhance the effects of GABA** in the brain.

Explain, with the aid of diagrams, from a physiological perspective (**depolarization, repolarization, and action potentials**, etc.), how this medication is likely to promote sleep in Ms. Aniya by focusing on the **GABAergic mechanism**.

Also, write on **transmission at the neuromuscular junction in myasthenia gravis**.

2. A researcher is investigating the neural basis of different behavioral states in human subjects. They administer various pharmacological agents and observe the resulting changes in physiological responses and reported experiences. **List any FIVE neurotransmitters and their effects on neurotransmission in various parts of the human body.**

3. A neuroscientist is studying the activity of a neuron in the brain of a research participant. They deliver a series of subthreshold excitatory stimuli to different neurons that synapse onto this target neuron. They also deliver a rapid sequence of subthreshold excitatory stimuli to a single presynaptic neuron that also synapses onto the same target neuron.

Explain how the target neuron might reach its threshold for firing an action potential in each of these scenarios, detailing the processes of **spatial summation** and **temporal summation** (with necessary labelled diagrams) in synaptic transmission.

Write on **action potentials in smooth muscles**.

4. An old man came to NTH with chest pain for a week.

Describe the **phases of the action potential** in a typical **ventricular contractile cell** and explain how these phases prevent the heart from developing **tetanic contractions**.

5. The **human heart** has four chambers, unlike an **octopus**, which has two hearts in its body.

Explain how the **prepotential (pacemaker potential)** in **SA node cells** contributes to the **automaticity of the heart**, and describe the **ionic changes** responsible for this phase.

SECTION E – Answer the Question Below (No Choice)

Marks = 3

1. **Draw a labelled diagram of the pump that resolves the problem caused by the Donnan Equilibrium in the cells.**

MUCH LOVE FROM DR OWEN  

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